

Evaluating Barriers to the Adoption of Delivery Drones in Rural Healthcare Supply Chains: Preparing the Healthcare System for the Future

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Abstract—Emerging technologies, such as delivery drones have enormous potential to revolutionize the rural healthcare supply chain (RHSC) and assist the countries in achieving sustainable development goal 3: good health and well-being. The advantages of delivery drones include faster delivery, high responsiveness, and a lack of dependence on road infrastructure. Given these advantages, prior literature on drone delivery has extensively explored their applicability in last-mile delivery from the e-commerce or retailer's perspective. However, their applicability in RHSC from an academic as well as practical perspective is still at a nascent stage. The present study fills this gap by examining the barriers inhibiting the adoption of delivery drones in RHSC using the Grey Decision-Making Trial and Evaluation Laboratory technique. The results suggest that “lack of government regulations” is the most critical barrier. The study findings indicate that seven barriers are causal, and six are the effect barriers. “Limited load carrying capacity,” “low flight range,” and “difficulty in flying in adverse weather conditions” are the most prominent cause barriers. Moreover, “lack of skilled manpower,” “limited accuracy of navigation system,” and “lack of leadership commitment” are the most prominent effect barriers. We also perform sensitivity analysis to prove the robustness of the study output. The study results provide an in-depth understanding to the decision-makers and policymakers to help pave the way for the successful adoption of delivery drones in RHSC.

Index Terms—Barriers, delivery drones, Grey Decision-Making Trial and Evaluation Laboratory (grey-DEMATEL), rural healthcare supply chain (RHSC), sustainable development goals (SDGs).

I. INTRODUCTION

THE sustainable development goals (SDGs), adopted by the United Nations Member States in 2015, focus on balancing economic, environmental, and social development to ensure that people enjoy prosperity and live in harmony with nature. By the

vow to leave no one behind, SDG 3 focuses on good health and well-being for all by 2030. It also aspires to ensure universal health coverage and access to vaccines and medicines for all [1]. In the past years, incredible progress has been made against the various causes of deaths, leading to reduced child mortality and maternal mortality rate and increased life expectancy [2]. Despite this remarkable progress, half of the world's population lacks access to essential healthcare services [3]. In developing countries, a large population still lives in rural areas which are remotely located and have an ill healthcare system. The rural healthcare facilities (RHF) often lack resources such as medicines and blood as the daily demand is on a walk-in basis which cannot be predicted [4]. The timely distribution of medicines and vaccines to these areas is also difficult because of the transportation challenges, such as lack of road infrastructure and inaccessible terrain [5].

Moreover, disasters, such as the outbreak of the COVID-19 pandemic could further aggravate the situation because the healthcare supply chains have to deal simultaneously with the regular healthcare issues and the disaster [6]. For example, in rural areas of Bihar, India, the last-mile delivery of medical essentials is carried out by healthcare workers, who were assigned COVID-19 related duties. This impacted the distribution of medicines to the rural population [7]. Furthermore, the transportation and delivery of medicines and supplies, such as personal protective equipment (PPE) kits, gloves, masks, and alcohol-based hand sanitizer, to confine the spread of the pandemic becomes challenging and time-consuming due to poor infrastructure [8], [9]. Delivery drones can be a viable option to overcome these challenges [9], [10], especially given the high urgency, high value, and small size of the medical essentials, such as medicines and blood supplies [11].

Drones, also known as unmanned aerial vehicles (UAVs), are remotely piloted aerial vehicles that fly over tough terrain. As a result, they could be utilized to ensure the timely delivery of medicines and other necessities to RHF [5]. Faster delivery is the most talked-about advantage of UAVs [9]. Other benefits include high responsiveness, low environmental pollution, avoiding road congestion, ensuring on-demand delivery to health systems that cannot afford to store cold-chain products, and avoiding human labor [12], [13]. UAVs have already found their application in various sectors, such as emergency response [14], military operations [15], mining [16], agriculture [17], and environmental protection [18]. The idea of delivery using drones

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quickly became popular after Amazon revealed its Prime Air service to deliver products within 30 min using UAVs [19]. And the race to fill the sky was soon joined by DHL with its Parcelcopter, Google with its Project Wing, Walmart, and others [18], [20]. However, these projects are still at the development stage.

One real-life example of delivery drones in the healthcare supply chain is Zipline, which currently operates in east African countries, namely Ghana and Rwanda. The Muhanga distribution center of Zipline in Rwanda delivers approximately 1400 blood samples per day to rural health facilities (RHF). The delivery drones of Zipline are flat-wing high-speed drones that have reduced the delivery of medical supplies from 4 h to less than 30 min [11]. Moreover, in Ghana, Zipline joined the war against COVID-19 by flying COVID-19 test samples from 1000 RHF to laboratories. RHF, in particular, send COVID-19 samples to Zipline distribution centers, which then deliver these samples to testing facilities using delivery drones [21]. Thus, reducing the test samples delivery time from hours to under an hour in some cases [9]. Boudway [22] suggested that the delivery drones could bypass the need for freezers to store COVID-19 vaccines by delivering small doses to rural areas as demand arises. Despite the various advantages, the use of delivery drones in rural healthcare supply chains (RHSC) is still nascent, highlighting the need to understand the factors obstructing their adoption in RHSC. Scholars also suggest that studying factors obstructing technology adoption helps identify and negate critical barriers that may impede the adoption of technology [23], [24], [25].

The literature acknowledges the use of drones as a part of successful delivery strategy. However, these studies majorly focus on product delivery using UAVs in business-to-customer settings, i.e., for last-mile delivery from the e-commerce or retailer's perspective. For example, Agatz et al. [26] and Carlsson and Song [18] proposed a combined truck–drone traveling salesman problem for fast and cost-efficient home delivery of packages. Baloch and Gzara [19] proposed a UAV-based parcel delivery network for e-retailers under competition. Dayarian et al. [27] proposed a drone resupply network for faster home delivery where drones resupply trucks. Raj and Sah [28] analyzed critical success factors (CSFs) for drones in the logistics sector, and Sah et al. [29] studied barriers to drone logistics in the last-mile delivery of e-commerce or food products. However, the strategic field of action to improve the adoption of drones which encompass addressing the barriers, especially in RHSC context, is a pertinent issue which has not been addressed.

The use of drones in the RHSC context differs from that of e-commerce or retail deliveries because in RHSC, generally, a hub-and-spoke model is used. Here, a hub is a distribution center that stores medical supplies, and the RHF are spokes. Generally, each rural area has one healthcare facility and is served by the hub. The distribution center also has a fleet of drones that deliver medical supplies to RHF as per demand. Moreover, the RHF are generally located far away from distribution centers and each other [12]. Contrarily, in e-commerce or retail deliveries, a combined truck and drone logistics system is more common and popular among transportation scientists [11]. In this case, the demand can be considered clusters, which are generally in the same city. Moreover, e-commerce or retail

drone deliveries are suitable in urban areas because it offers more business to compensate for the cost of delivery drone adoption. However, this is not the case with rural areas. Rural people have a limited spending power, posing serious threat to the economic viability of the drone delivery system, which may ultimately affect leaders' intention to adopt the technology. Moreover, the factors like lack of skilled workforce, and location of rural areas near sensitive locations of national importance, make drone-based delivery in RHSC an important topic to explore. Precisely, the above discussion highlights the need to deepen the understanding of potential barriers and their causal relationships for delivery drone adoption in the RHSC context.

Therefore, to accelerate the adoption of delivery drones in RHSC and achieve SDG 3, this study proposes the following research questions.

RQ1. What are the key barriers inhibiting the adoption of delivery drones in RHSCs, and how to prioritize them?

RQ2. How to categorize the identified barriers into cause-and-effect groups and uncover the causal relationship between them for the successful adoption of delivery drones in RHSCs?

The study adopts a multicriteria decision making (MCDM) approach, to address these research questions. Precisely, the study uses a combination of grey-systems theory and Decision-Making Trial and Evaluation Laboratory (DEMATEL), namely, Grey-DEMATEL, which helps identify the degree of influence of each barrier on other barriers. Grey-systems theory assists in capturing vagueness in expert judgment, providing improved analysis compared to other MCDM methods [30], [31], [32]. Under this method, first, an extensive literature survey followed by expert opinion is conducted to identify barriers to the use of delivery drones in RHSC. Second, Grey Decision-Making Trial and Evaluation Laboratory (Grey-DEMATEL) is utilized to prioritize the barriers, categorize them into cause and effect groups, and assess the causal relationship between them [30]. The results show that a “lack of government regulations” (GRs) is the most prominent barrier hindering delivery drones' adoption in RHSC. The results also show that seven barriers including “limited load carrying capacity” (LCC), “low flight range” (FR), and “difficulty in flying in adverse weather conditions” (AWCs) are the most prominent cause barriers. Six barriers including “lack of skilled manpower” (SM), “limited accuracy of navigation system” (NS), and “lack of leadership commitment” (LC) are the most prominent effect barriers.

The remainder of the study is structured as follows. **Section II** presents relevant literature on delivery drones to highlight research gaps. **Section III** presents the methodology that describes the research framework and the Grey-DEMATEL technique used for analysis. **Section IV** presents the application of the proposed model. **Section V** presents a detailed discussion of the study findings. Finally, the study concludes in **Section VI**.

II. LITERATURE REVIEW

A. Delivery Drones for Disaster Response and Healthcare Supply Chains

The first relevant literature stream is the use of drones in disaster and the healthcare supply chain. In this context, Haidari

et al. [33] assessed the impact of drones on vaccine distribution using simulation modeling. Their results show that using drones for vaccine distribution can significantly improve vaccine availability and reduce logistics costs. Dhote and Limbourg [12] proposed four location models for locating the drone launch bases for biomedical material supply. Using a real-life case of Brussels, the authors show that charging stations and drone sharing among the hospitals are necessary for improving the range, increasing market share, and reducing costs. Chowdhury et al. [34] studied the distribution center location problem for disaster response, where the distribution centers use delivery drones to transport emergency commodities. The analysis using three real-life cases highlights the influence of drone speed, flight height, inventory, and reordering cost on the system's performance. Amukele [35] performed experiments on delivery drones in healthcare supply chains and reported that laboratory specimens are not affected if the flight time is below 40 min. Jayabalan et al. [36] performed a qualitative study to identify ethics, challenges, and opportunities for drone delivery in the healthcare supply chain from the program manager and field staff perspectives. Cawthorne and Wynsberghe [37] proposed an ethical framework to facilitate awareness of ethical issues related to drones in the healthcare supply chain. Hiebert et al. [38] provide a review of the application of drones in healthcare and related sectors. While some of these existing studies discuss challenges to drone adoption in healthcare supply chain, they lack in terms of a comprehensive approach to decision making. Unlike these existing studies, the present study explicitly focuses on comprehensively examining barriers obstructing the adoption of delivery drones in RHSC context.

B. Literature on Drone Delivery

The second relevant literature stream is on drone-based delivery, which suggests that the transportation scientists have majorly proposed models for last-mile drone-based delivery and relied on quantitative methods and tools for analyzing them. For example, Dorling et al. [39] presented two vehicle routing problems for drone-based delivery. Their first problem minimizes cost with a constraint on delivery time, and the second problem minimizes delivery time within a given budget. The models were solved using simulated annealing. Agatz et al. [26] proposed an integer programming model for truck–drone collaborated logistics to improve last-mile delivery. Their results show significant cost savings in truck–drone delivery as compared to truck-only delivery. Similarly, Carlsson and Song [18] presented a truck–drone collaborated logistics, in which the truck itself is moving to provide package delivery service. Dayarian et al. [27] proposed a new variant of truck–drone coordinated logistics, especially for urban and dense settings. They propose that the trucks provide regular last-mile delivery, and the delivery drones fulfill the demand of trucks. Baloch and Gzara [19] designed a distribution network under competition to assess the economic desirability of drone-based delivery. Their findings suggest that, currently, drone-based delivery is difficult in urban areas or densely populated areas. With GRs and proper technology, the service provider can target price-sensitive

customers by providing discounts. Perera et al. [40] examined the impact of drones on retail deliveries. They reported that retailers should partition the market into delivery zones and offer the best possible delivery to each zone using delivery drones. Readers may refer to Moshref-Javadi and Winkenbach [11] and Chung et al. [41] for an extensive review on drone-based logistics.

Raj and Sah [28] examined CSFs that play a significant role in facilitating the adoption of delivery drones for improving customer service and gaining a competitive advantage through faster products delivery. Sah et al. [29] studied why the adoption of delivery drones is low in last-mile delivery, despite their ability to reduce cost and improve responsiveness of last-mile operations. They find that “accountability of drone owners” followed by “insurance obligations” is the most prominent barriers that should be resolved on priority to accelerate delivery drone's adoption in last-mile delivery. The prowess of delivery drones is not affected by just a few factors. Rather, it is afflicted by a multitude of barriers that often cut across organizational boundaries. In addition, a nuanced understanding of barriers to adoption of delivery drones in RHSC requires integration of interdisciplinary knowledge which spans across domains, such as technology, management, and healthcare among others. While Sah et al. [29] discuss the barriers; they do not touch upon some important RHSC specific barriers. For example, the authors do not include LC which refers to leaders' awareness of technology which molds their perception and willingness to adopt it [42]. The industry experts, during our interaction with them, revealed that in case of last-mile delivery, the service provider may charge consumer extra for the fast delivery performed using drones, thus compensating for the cost of implementation. However, in RHSC context, the rural people have very limited spending power, which deters leaders from adopting delivery drones. Another important barrier highlighted by the industry experts is the “security concerns” (SCs). They argue that many times the rural areas are nearby international borders, army bases, or other sensitive areas. The delivery drone may have to fly over or around these sensitive locations to reach these areas, posing a serious threat to the national security. While, the existing literature studies challenges to delivery drone adoption, they lack in terms of comprehensively assessing the impact of all the relevant barriers. The present study differs from the extant literature, as it not only comprehensively assesses the important barriers but also examines the plausible existence of a causal relationship between the barriers that are exclusive to RHSC.

III. METHODOLOGY

The research framework adopted in the present study to analyze the barriers inhibiting the adoption of delivery drones in RHSC is based on the DEMATEL method (Fig. 1). Like DEMATEL, various multiattribute decision-making (MADM) techniques exist in the literature, such as analytical hierarchy process, TOPSIS, and analytical network process [43]. However, these MADM techniques assume that criteria are independent of each other, which is unrealistic [30]. DEMATEL, unlike these MADM techniques, tries to identify the interdependence

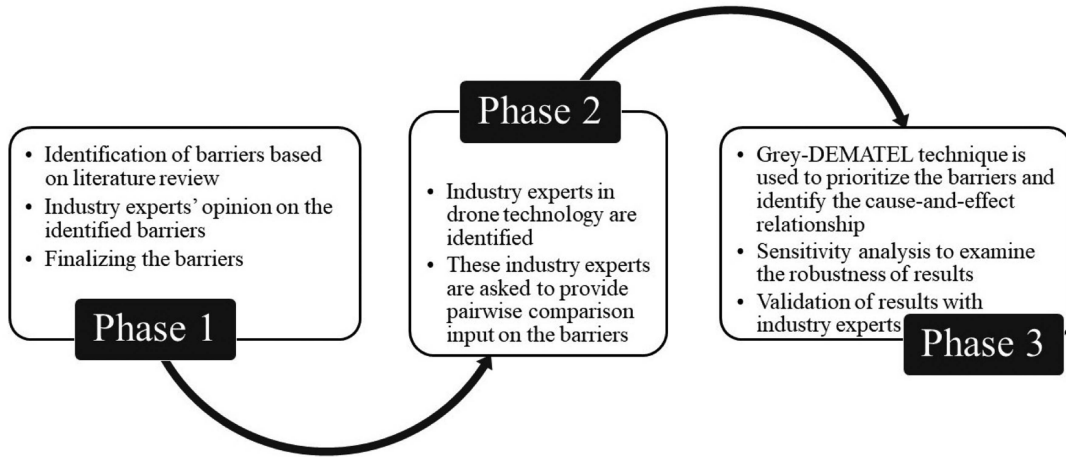


Fig. 1. Proposed framework.

between factors of interest in the complex structural model through a causal diagram [44], [45]. Interpretive structural modeling, another popular MADM method, also establishes causal relationships among factors, but it does not specify the severity of the effect, unlike DEMATEL [29], [47]. DEMATEL evaluates the interaction between the factors by categorizing them into cause-and-effect groups. The causal diagram obtained from DEMATEL presents the contextual relationship and the influence among the factors and severity of influence [30], [46]. Despite these advantages, DEMATEL has a limitation that it cannot account for the ambiguity in the expert inputs, affecting the accuracy of the results.

There exist two popular theories, namely grey theory and fuzzy theory that handle ambiguity or uncertainty in human judgments. Fuzzy theory, grounded in subjective uncertainty, suggests that every concept is abstract and there exists no common definition which is shared by everyone. Mathematically, the fuzzy theory focuses on the vagueness of the boundary between a set and its complement. Grey theory, on other hand, is based on objective uncertainty with a clear lower and upper limit and an unknown position within that limit [47]. The experts in our study have years of experience in this field and therefore are aware of the extent of relationship between the barriers. However, there is always uncertainty regarding the exact strength of relationship within the limit/range. Therefore, we use grey theory and combine it with DEMATEL (Grey-DEMATEL) to capture ambiguity and improve the accuracy of the results [45]. The use of Grey-DEMATEL is also in line with the past literature that has successfully used it to assess critical factors in various contexts, such as Industry 4.0 [48], closed-loop supply chain [32], supplier/technology provider selection [49], frugal innovation [50], and green business failure [45].

The proposed framework has three phases (Fig. 1).

Phase 1: In this phase, the barriers inhibiting the adoption of delivery drones in RHSC are identified based on an extant literature review followed by expert opinion. For this purpose, we first searched for research papers on delivery drones using the keywords “drone,” “UAV,” “delivery,” “rural,” “healthcare,” and “barriers” in various combinations. After eliminating duplicate

and irrelevant articles, we manually searched the remaining articles for barriers obstructing the adoption of delivery drones and identified 16 barriers. We then contacted five experts working in the field of healthcare products delivery using drones. Among these experts, three agreed to review the identified barriers. All the 16 barriers were discussed with the experts through an online semistructured interview. During the interview, the experts were free to add, remove, or modify any barrier. This discussion and feedback resulted in 13 barriers that were found relevant for the adoption of delivery drones in RHSCs. This step also helped us ensure we did not miss any relevant barriers. The finalized barriers and their description are present in Table I.

Phase 2: The objective of the second phase is to gather data from relevant sources for analysis. For this purpose, we initially identified and contacted 30 industry experts currently working on drone-based delivery and drone policymaking. These experts were reached over professional networking websites due to COVID-19. Among the contacted experts, 17 agreed to participate, while only 10 actually participated. Details of the expert profile are present in Table II. The industry experts were asked to provide pairwise comparison inputs on the finalized barriers using the linguistic scale, shown in Table III [32].

Phase 3: In this phase, expert inputs are evaluated using the Grey-DEMATEL technique to categorize the barriers and uncover the causal relationship between them. We also perform sensitivity analysis to confirm the robustness of our results. Finally, the results of the study are validated based on the discussion with the industry experts.

The Grey-DEMATEL procedure adopted in this study can be summarized in seven steps, which are as follows.

Step 1: Let n be the number of barriers identified for the analysis, and p be the number of experts. Each expert is asked to evaluate the influence of barrier i over barrier j , where $i, j = 1 \dots n$ and $i \neq j$. Hence, p initial-relation matrices are obtained. Let $\otimes k_{ij}^p$ be the grey number representing the influence of barrier i over j of expert p . Then $\otimes K_{ij}^p$ represents its lower limit and $\bar{\otimes} k_{ij}^p$ represents its upper limit, i.e., $\otimes k_{ij}^p = (\otimes k_{ij}^p, \bar{\otimes} k_{ij}^p)$.

TABLE I
BARRIERS AND DESCRIPTION

Code	Barrier	Description	Source
GR	Lack of government regulations	Unclear regulations and low intervention of government in promoting and supporting delivery drone technology.	[51], [52], [53]
LC	Lack of leadership commitment	Leaders are hesitant to promote and use drone technology due to perceived operational complications, lack of knowledge, and reluctance to change the existing system.	[28]
TP	Threat to privacy	Drones have audio–visual recording devices which may interrupt the general public's privacy during drone operations.	[29], [51], [54]
AWC	Difficulty in flying in Adverse weather conditions	Drones are sensitive to extreme weather conditions such as rain, snow, and wind. Thus, impacting drone operations and efficiency.	[11], [55], [56]
IDC	High infrastructure requirement and development cost	Implementation of drones and their smooth operations need robust infrastructures such as launch and charging stations. And developing infrastructure is a costly process.	[33], [51], [57]
FR	Low flight range	Drones have a restricted range of flight because of battery size and capacity, which is likely to impact long-distance deliveries.	[10], [65]
LCC	Limited load carrying capacity	Drones are not capable of transporting hefty products.	[39], [58]
CC	Collision and crashing	The risk of the drone colliding with other drones or objects during operations may cause economic loss and damage to the drone. Moreover, drone's crashing endangers people's property and their health.	[41], [59]
NS	Limited accuracy of navigation system	The limited accuracy of GPS technology leads to product delivery at the wrong destination.	[11]
ND	Noise disturbance	The buzzing sound of drones may cause noise pollution, which may be bothersome to nearby residents.	[41], [60]
CH	Threat of cyber-attack and hacking	Drones are vulnerable to cyber-attacks and hacking, which may lead to product and drone theft. Hacked drones may also be used for mischievous activities.	[61], [62]
SM	Lack of skilled manpower	Drones use sophisticated tools and techniques such as machine learning and artificial intelligence; hence skilled workforce is needed. The availability of a workforce is influenced by the fact that the industry is in a nascent stage, and therefore there are very few drone institutes and training centers.	[57]
SC	Security concerns	Use of drones for criminal or terrorist activities. Moreover, flying drones over parliament buildings, military bases, international borders, and other sensitive areas is likely to threaten national security.	[63], Expert opinion

Step 2: The p initial-relation matrices are merged to obtain average grey-relation matrix A using the below-mentioned equations

$$\otimes k_{ij} = \left(\frac{\sum_p \otimes k_{ij}^p}{p}, \frac{\sum_p \bar{\otimes} k_{ij}^p}{p} \right) \quad (1)$$

$$A = \begin{bmatrix} \otimes k_{11} & \cdots & \otimes k_{1n} \\ \vdots & \ddots & \vdots \\ \otimes k_{n1} & \cdots & \otimes k_{nn} \end{bmatrix}. \quad (2)$$

Step 3: The average grey-relation matrix A is used to compute crisp relation matrix C . This computation involves three steps, namely, normalization of grey values [see (3)–(5)], calculating total normalized crisp values [see (6)], and obtaining final crisp values and crisp relation matrix [see (7) and (8)].

$$\Delta_{\min}^{\max} = \max_j \bar{\otimes} k_{ij} - \min_j \underline{\otimes} k_{ij} \quad (3)$$

$$\underline{\otimes} \tilde{k}_{ij} = \left(\underline{\otimes} k_{ij} - \min_j \underline{\otimes} k_{ij} \right) / \Delta_{\min}^{\max} \quad (4)$$

$$\bar{\otimes} \tilde{k}_{ij} = \left(\bar{\otimes} k_{ij} - \max_j \bar{\otimes} k_{ij} \right) / \Delta_{\min}^{\max} \quad (5)$$

where $\underline{\otimes} \tilde{k}_{ij}$ and $\bar{\otimes} \tilde{k}_{ij}$ are normalized lower and upper values of the grey number $\otimes \tilde{k}_{ij}$, respectively.

$$l_{ij} = \left(\left(\underline{\otimes} \tilde{k}_{ij} \times \left(1 - \underline{\otimes} \tilde{k}_{ij} \right) \right) + \left(\bar{\otimes} \tilde{k}_{ij} \times \bar{\otimes} \tilde{k}_{ij} \right) \right) / \times \left(1 - \underline{\otimes} \tilde{k}_{ij} + \bar{\otimes} \tilde{k}_{ij} \right) \quad (6)$$

$$\hat{l}_{ij} = \min_j \underline{\otimes} k_{ij} + (l_{ij} \times \Delta_{\min}^{\max}) \quad (7)$$

$$C = \left[\hat{l}_{ij} \right]. \quad (8)$$

Step 4: The crisp relation matrix C is normalized using a normalization factor N [see (9)] to obtain a normalized direct crisp relation matrix D [see (10)]

$$N = 1 / \max_{1 \leq i \leq n} \sum_{j=1}^n \hat{l}_{ij} \quad (9)$$

$$D = N \times C. \quad (10)$$

Step 5: Total-relation matrix T is calculated from normalized direct crisp relation matrix D and identity matrix I , as shown in the following:

$$T = D \times (I - D)^{-1}. \quad (11)$$

Step 6: In this step, row sums R_i and column sums C_j are computed using total-relation matrix T [see (12) and (13)]

$$R_i = \sum_{j=1}^n m_{ij} \quad \forall i \quad (12)$$

TABLE II
DETAILED PROFILE OF EXPERTS

Expert	Work experience	Industry/academia	Designation/position
E1	More than 10 years	Industry	Vice President Business Development (VP)
E2	More than 5 years	Industry	Senior UAV Trainer
E3	More than 10 years	Industry	Chief Executive Officer (CEO)
E4	More than 5 years	Industry	Assistant General Manager (AGM)
E5	More than 10 years	Industry	Drone Lead
E6	More than 5 years	Industry	Senior Engineer
E7	More than 5 years	Industry	Senior Engineer
E8	More than 5 years	Industry	Chief Technology Officer (CTO)
E9	More than 5 years	Industry	Business Development
E10	More than 10 years	Industry	Chief Operating Officer (COO)

TABLE III
LINGUISTIC SCALE AND CORRESPONDING GREY VALUES

Terms/intensity of influence	Linguistic scale	Grey scale
No influence	N	(0,0)
Very low influence	VL	(0,0.25)
Low influence	L	(0.25,0.50)
High influence	H	(0.50,0.75)
Very high influence	VH	(0.75,1.00)

$$C_j = \sum_{i=1}^n m_{ij} \quad \forall j \quad (13)$$

where m_{ij} represents the elements of the total-relation matrix T . The total effect given and received by a barrier i (also referred to as prioritization or degree of importance) is given by $(R_i + C_j)$. Moreover, $(R_i - C_j)$ refers to the net effect contribution of the barrier i to the entire system. A positive value of $(R_i - C_j)$ indicates that the barrier i is a cause, whereas a negative value indicates that it is an effect.

Step 7: Final step is to compute a threshold value to plot the digraph. Threshold value θ is set to avoid negligibly small effects in the digraph. Threshold value θ is usually calculated as the sum of the mean μ and standard deviation σ of the elements of the total-relation matrix T [47]

$$\theta = \mu + \sigma. \quad (14)$$

If $\theta \leq m_{ij}$, then barrier i causes barrier j . Moreover, the coordinates of barriers for the cause-and-effect digraph are $((R_i + C_j), (R_i - C_j)) \forall i = j$.

IV. APPLICATION OF THE PROPOSED FRAMEWORK

The proposed framework is practically tested in India because most of the Indian population still lives in rural areas and suffers a healthcare service shortage [13], [64]. Although substantial improvements have been made in the recent past, the Indian

RHSC is still developing. The rural areas of India still face a shortage of medical essentials. Poor road infrastructure and challenging geography further aggravate this situation resulting in an inadequate supply of medicines and vaccines [64], ultimately affecting the country's objective of achieving SDG 3 good health and well-being for all by 2030. Therefore, it is essential to embrace delivery drones in RHSC to assist in faster delivery of medical essentials hard to reach locations [13]. This study aims to accelerate delivery drones' adoption in RHSCs by shedding light on the barriers and assisting in strategizing mitigation strategies.

In this vein, we gather data from industry experts engaged in the drone delivery sector to prioritize and find a causal relationship between the barriers. First, based on a comprehensive literature survey and industry expert opinion (detailed in Section III), 13 barriers affecting the adoption of delivery drones in RHSC in India were identified (Table I). Then the industry experts were asked to provide a pairwise comparison of the identified barriers. A total of 10 industry experts participated in the second phase of analysis; this is in line with the past literature [32], [45]. One expert is from the international body and is working on the use of drones in RHSC. Five experts are senior-level managers in the drone sector, such as CEO, CTO, COO, VP, and AGM, and work closely with regulatory bodies to make the Indian government's "Medicine from the Sky" project a success. One expert is associated with a firm engaged in delivering medicines, blood, and vaccines using drones. And three experts are senior level engineers in the drone sector. A brief expert profile is

TABLE IV
CRISP RELATION MATRIX

	GR	LC	TP	AWC	IDC	FR	LCC	CC	NS	ND	CH	SM	SC
GR	0.000	0.652	0.694	0.240	0.619	0.544	0.415	0.760	0.455	0.305	0.592	0.619	0.838
LC	0.599	0.000	0.433	0.315	0.626	0.329	0.324	0.471	0.317	0.293	0.387	0.576	0.504
TP	0.612	0.410	0.000	0.197	0.338	0.304	0.150	0.296	0.363	0.222	0.541	0.506	0.639
AWC	0.127	0.166	0.095	0.000	0.280	0.575	0.537	0.744	0.548	0.222	0.183	0.466	0.432
IDC	0.599	0.748	0.344	0.256	0.000	0.509	0.371	0.246	0.333	0.206	0.266	0.398	0.416
FR	0.751	0.455	0.344	0.438	0.518	0.000	0.733	0.590	0.661	0.144	0.219	0.556	0.416
LCC	0.702	0.531	0.160	0.519	0.428	0.688	0.000	0.590	0.439	0.240	0.223	0.488	0.374
CC	0.642	0.561	0.433	0.539	0.463	0.405	0.356	0.000	0.548	0.254	0.295	0.537	0.733
NS	0.289	0.425	0.269	0.285	0.277	0.452	0.197	0.440	0.000	0.114	0.147	0.398	0.374
ND	0.202	0.253	0.188	0.256	0.247	0.218	0.326	0.267	0.190	0.000	0.147	0.338	0.276
CH	0.642	0.501	0.602	0.139	0.323	0.161	0.130	0.217	0.231	0.175	0.000	0.428	0.673
SM	0.373	0.561	0.328	0.197	0.414	0.301	0.608	0.428	0.289	0.273	0.295	0.000	0.461
SC	0.872	0.701	0.714	0.254	0.338	0.346	0.244	0.500	0.548	0.240	0.431	0.537	0.000

TABLE V
NORMALIZED DIRECT CRISP RELATION MATRIX

	GR	LC	TP	AWC	IDC	FR	LCC	CC	NS	ND	CH	SM	SC
GR	0.000	0.097	0.103	0.036	0.092	0.081	0.062	0.113	0.068	0.045	0.088	0.092	0.124
LC	0.089	0.000	0.064	0.047	0.093	0.049	0.048	0.070	0.047	0.044	0.057	0.086	0.075
TP	0.091	0.061	0.000	0.029	0.050	0.045	0.022	0.044	0.054	0.033	0.080	0.075	0.095
AWC	0.019	0.025	0.014	0.000	0.042	0.085	0.080	0.110	0.081	0.033	0.027	0.069	0.064
IDC	0.089	0.111	0.051	0.038	0.000	0.076	0.055	0.036	0.049	0.031	0.039	0.059	0.062
FR	0.111	0.068	0.051	0.065	0.077	0.000	0.109	0.088	0.098	0.021	0.033	0.083	0.062
LCC	0.104	0.079	0.024	0.077	0.064	0.102	0.000	0.088	0.065	0.036	0.033	0.073	0.056
CC	0.095	0.083	0.064	0.080	0.069	0.060	0.053	0.000	0.081	0.038	0.044	0.080	0.109
NS	0.043	0.063	0.040	0.042	0.041	0.067	0.029	0.065	0.000	0.017	0.022	0.059	0.056
ND	0.030	0.038	0.028	0.038	0.037	0.032	0.048	0.040	0.028	0.000	0.022	0.050	0.041
CH	0.095	0.074	0.089	0.021	0.048	0.024	0.019	0.032	0.034	0.026	0.000	0.064	0.100
SM	0.055	0.083	0.049	0.029	0.062	0.045	0.090	0.064	0.043	0.040	0.044	0.000	0.068
SC	0.130	0.104	0.106	0.038	0.050	0.051	0.036	0.074	0.081	0.036	0.064	0.080	0.000

present in Table II. The data gathered from these experts resulted in 10 (13×13) initial relation matrices.

These matrices are aggregated using (1) and (2), and all the experts are assigned equal weight for homogeneity in judgment. This resulted in a single matrix, i.e., average grey relation matrix, which is employed to compute crisp relation matrix C (Table IV) using (3)–(8). The crisp relation matrix C is then normalized to obtain normalized direct crisp relation matrix D , as shown in Table V. The total-relation matrix T is then obtained using (11) (Table VI). The column sum C_j and row sum R_i are computed from the total-relation matrix T based on (12) and (13). $(R_i + C_j)$ and $(R_i - C_j)$ are employed to obtain the degree of importance of the barrier and its contributed net effect, respectively (Table VII). Finally, from (14), the threshold value $\theta = 0.222 + 0.067 = 0.289$ is set for the digraph.

V. RESULTS AND DISCUSSION

A. Prioritization of Barriers

Priority order of the identified barriers (RQ1) is calculated as $(R_i + C_j)$ where $i = j$, a higher value of $(R_i + C_j)$ indicates higher priority or importance of the barrier. The results show that “lack of GRs” has the highest priority among the identified barriers. This finding is in line with Raj and Sah [28] and suggests that the development of regulations for the civil use of drones could pave the way for the adoption of delivery drones, especially in RHSC. We find that SC has the second-highest priority, followed by “collision and crashing” (CC) (Table VII). This finding is in contradiction with Sah et al. [29], who report SC as among the least priority barriers. This difference in findings is due to difference in application area. Sah et al.

TABLE VI
TOTAL RELATION MATRIX

	GR	LC	TP	AWC	IDC	FR	LCC	CC	NS	ND	CH	SM	SC
GR	0.301	0.370	0.320	0.202	0.315	0.294	0.258	0.355	0.291	0.171	0.264	0.354	0.397
LC	0.314	0.220	0.235	0.175	0.265	0.219	0.203	0.263	0.222	0.142	0.196	0.288	0.291
TP	0.292	0.254	0.160	0.143	0.208	0.195	0.161	0.219	0.208	0.121	0.204	0.257	0.287
AWC	0.217	0.209	0.159	0.116	0.191	0.229	0.212	0.273	0.230	0.116	0.142	0.242	0.245
IDC	0.297	0.304	0.210	0.158	0.168	0.231	0.199	0.221	0.211	0.122	0.170	0.250	0.262
FR	0.363	0.311	0.243	0.212	0.276	0.200	0.280	0.310	0.293	0.134	0.191	0.314	0.308
LCC	0.338	0.303	0.206	0.213	0.251	0.279	0.171	0.296	0.251	0.140	0.180	0.290	0.285
CC	0.343	0.319	0.253	0.219	0.263	0.249	0.225	0.223	0.274	0.147	0.199	0.307	0.344
NS	0.210	0.218	0.164	0.137	0.171	0.189	0.146	0.208	0.131	0.090	0.124	0.209	0.213
ND	0.160	0.160	0.124	0.112	0.137	0.130	0.137	0.152	0.129	0.057	0.101	0.166	0.164
CH	0.284	0.254	0.234	0.127	0.196	0.166	0.148	0.197	0.180	0.110	0.123	0.236	0.280
SM	0.259	0.271	0.199	0.146	0.217	0.197	0.223	0.236	0.198	0.128	0.167	0.185	0.258
SC	0.373	0.337	0.293	0.179	0.248	0.238	0.207	0.289	0.271	0.145	0.220	0.307	0.248

TABLE VII
PROMINENCE AND NET EFFECT VALUES

Barriers	R_i	C_j	$R_i + C_j$	$R_i - C_j$	Identity
GR	3.892	3.752	7.644	0.140	Cause
LC	3.033	3.528	6.561	-0.495	Effect
TP	2.709	2.800	5.509	-0.091	Effect
AWC	2.581	2.138	4.719	0.442	Cause
IDC	2.803	2.905	5.709	-0.102	Effect
FR	3.436	2.816	6.252	0.619	Cause
LCC	3.202	2.569	5.771	0.632	Cause
CC	3.365	3.241	6.606	0.124	Cause
NS	2.210	2.889	5.099	-0.679	Effect
ND	1.729	1.623	3.352	0.106	Cause
CH	2.533	2.281	4.814	0.252	Cause
SM	2.683	3.406	6.089	-0.723	Effect
SC	3.356	3.583	6.939	-0.227	Effect

[29] worked on the delivery drones' adoption in urban areas for e-commerce or retail deliveries. The urban areas are located far from sensitive locations, such as parliament buildings, military bases, and international borders. Hence, the SCs are negligible. However, in the case of RHSCs, the delivery drones have to travel a long distance and sometimes the rural areas can be close to international borders or military bases. The delivery drone may have to fly over or around these sensitive areas for faster deliveries. Hence, the security concern is high in RHSC context. In this regard, UAV traffic management (UTM) platforms could potentially eliminate the risk of drones flying over sensitive air spaces. The UTM may provide information regarding sensitive air spaces and direct the drones to alternate path or air space, thus reducing security risk. The UTM platforms can also help mitigate the third most important barrier CC, by providing real-time information of UAVs operating in the same airspace. As per the Press Bureau Information [20], the Indian government

has initiated identifying airspaces for civil use and has developed a comprehensive UAV management platform, currently under test, named Digital Sky Platform. This platform will provide interactive airspace maps with information on fly-zones and no-fly-zones for the UAVs to accelerate the adoption of drones in various sectors.

B. Cause and Effect Groups and the Relationship Between Barriers

Based on study results, this section provides details on cause-and-effect groups and the relationship between the barriers (RQ2). Barriers having positive ($R_i - C_j$) values are categorized as the cause, as their influence on other barriers (R_i) is more than the influence received from those barriers (C_j). A higher value of ($R_i - C_j$) indicates a greater influence of the barrier on the system. Findings suggest that LCC is the most

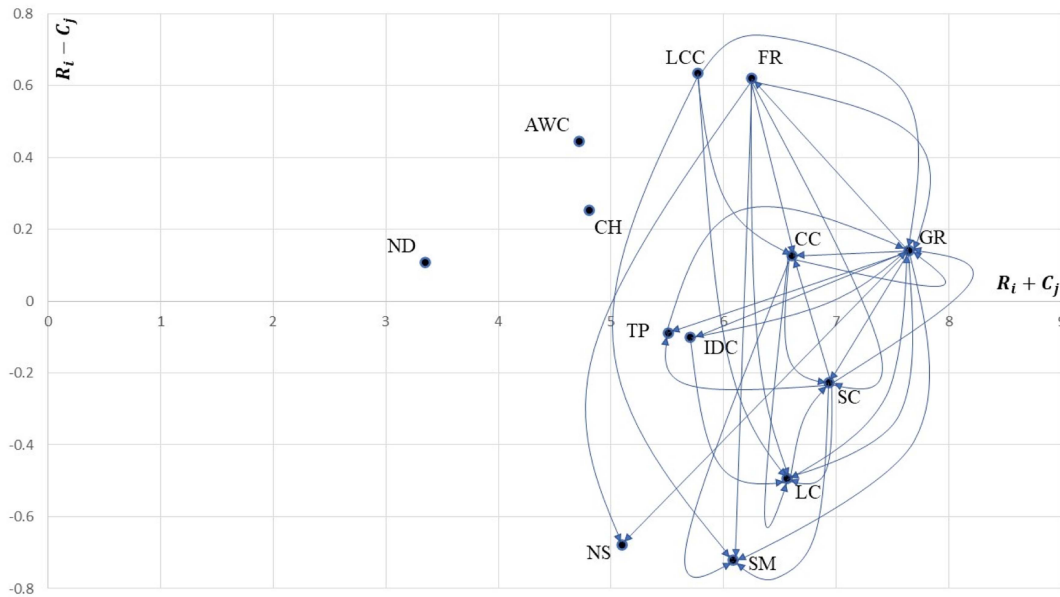


Fig. 2. Cause-and-effect relationship digraph.

important barrier in the cause group, closely followed by “low FR,” and “difficulty in flying in AWC.” Other barriers identified under cause group in order of their influence on other barriers are “threat of cyber-attack and hacking” (CH), “lack of GRs,” CC, and “noise disturbance” (ND), i.e., $LCC > FR > AWC > CH > GR > CC > ND$ (Table VII). These findings are in line with [28], who find that technical aspects play a vital role in the successful adoption of product delivery drones. However, the authors consider technological aspects as one factor, limiting their ability to highlight the relative importance and interrelationship between the constituents of technical aspects.

In the RHSC context, higher load-carrying capacity is necessary to deliver heavy medical supplies, such as blood and frozen plasma. Sometimes medicines require low-temperature storage, and drones may have to carry extra weight to maintain the desired temperature during transportation. Moreover, in situations like COVID-19, higher load-carrying capacity drones could not only help in transporting masks and gloves in bulk to the RHF but also assist in transporting PPE kits. FR is the second most prominent cause barrier as the drones employed in RHSC have to make a round trip from the distribution center to the rural health facility [12]. The rural areas are generally located far from distribution centers and lack resources, including a skilled workforce and electricity to charge drone batteries. Therefore, if the FR is less, the drone may not return to the distribution center. Similarly, other barriers under cause group, namely “difficulty in flying in AWC,” “threat of CH,” CC, and ND are related to technical aspects of delivery drones [29]. These findings suggest that the delivery drone technology is still in a nascent stage. We suggest that the decision-makers should focus on technological upgradation in the following order $LCC > FR > AWC > CH > CC > ND$ to ensure smooth adoption of delivery drones in RHSC. Some of our suggestions to address these causal barriers include 1) locating charging stations at RHF to compensate for the low FR; 2) decision-makers may explore the possibility of multiple drones carrying a single heavy product to compensate

for low load-carrying capacity; 3) fixed-wing vertical take-off and landing (VTOL) drones could be adopted to overcome the difficulty of flying in AWCs. The fixed-wing VTOLs may also reduce noise pollution because they have small rotor blades which generate low noise compared to quadcopters.

The study results show that “lack of GRs” was categorized as a cause barrier but is also affected by “lack of LC,” “threat to privacy” (TP), “high infrastructure requirement and development cost” (IDC), “low FR,” LCC, CC, and SC (Fig. 2). These findings are interesting and in sharp contrast with the existing literature because it is uncommon for a cause group barrier to be affected by approximately 50% barriers. For example, Raj and Sah [28] report that GR is a cause and is not influenced by any other barrier. A possible explanation for such findings could be that the government is likely to support delivery drone technology in RHSC if the healthcare sector leaders have a positive perception about it and the technology is promising. Moreover, GR’s categorization under cause group also suggests that policymakers should extend support to the delivery-drone technology in the form of national regulations and policies. For example, the government should smoothen the approvals process for trial and testing, improve access to relevant materials and technologies, and incentivize drone manufacturers [55]. While the results show a recursive relation between various barriers and GR, the GR’s categorization under cause group suggests that the policymakers need to take the first step and promote drone-based delivery in RHSC. However, later, the process becomes iterative with continuous flow of relevant ideas and information between the manufacturer and the policymakers. Such actions not only assist in strengthening GRs but could also help in the upgradation of delivery drones technology [52].

The effect group consists of barriers whose influence on other barriers (R_i) is less than the influence received from those barriers (C_j), i.e., $(R_i - C_j)$ is negative. The results show that “lack of SM,” “limited accuracy of NS,” “lack of LC,” SC, “high IDC,” and TP fall under effect group (Table VII). Such findings

TABLE VIII
EXPERT GROUP AND SCENARIO DETAILS

Group	Expert	Scenarios			
		<i>Scenario 1</i>	<i>Scenario 2</i>	<i>Scenario 3</i>	<i>Scenario 4</i>
Group 1 (G1)	E2, E4, E6, E7, E8, E9				
Group 2 (G2)	E1, E3, E5, E10	$G2 = 0.75 * G1$	$G2 = 0.95 * G1$	$G2 = 1.15 * G1$	$G2 = 1.35 * G1$

suggest that, even though SM, NS, LC, SC, IDC, and TP are critical factors, they arise later in the adoption process when the technology is ready to be implemented. The categorization of SM, NS, SC, and TP are unique to our study.

As shown in the cause-and-effect digraph (Fig. 2), the barriers in the effect group are substantially influenced by multiple barriers. For example, LC is influenced by GR, IDC, FR, LCC, CC, and SC. These findings suggest that high cost associated with delivery drone adoption deters top management from adopting the technology. This is because, unlike urban last-mile delivery, recovering the cost of technology implementation in RHSC is difficult due to the rural population's low spending power. The manufacturers should work toward improving technology and gaining government support including subsidies to motivate the leaders to adopt the technology for societal good. "Lack of SM" is another important barrier that is influenced by multiple barriers (Fig. 2). The factors that substantially influence SM include GR, FR, LC, CC, and SC. The reason for such findings is that the technology is still in the nascent stage and struggling to be ubiquitous. Therefore, there are very few drone schools or training centers, and the supply of skilled workforce is less, ultimately affecting the availability of skilled workforce. However, as critical barriers are eliminated, and drone-based delivery becomes popular, demand for workers in other forms of transportation will decline, and people will begin to consider the drone industry as a potential career option [41]. Another important effect barrier is SC. Findings show that resolving technical barriers (namely, FR and CC) and formulating policies for delivery-drone operations could substantially eliminate SCs (Fig. 2). For example, the policymakers could identify airspaces that delivery drones could not use and communicate the same to the firms or UTM. Moreover, the regulations should only allow operations through UTM platforms to avoid CC and avoid drones breaching sensitive airspaces [20].

C. Sensitivity Analysis

Scholars suggest that in MADM techniques, the data collection process may induce vagueness and uncertainty in the study results. Hence, sensitivity analysis is used as a tool to eliminate this vagueness and assess the robustness of the results [31], [32], [48]. Under sensitivity analysis, the study parameters are varied within the expected range to understand how they affect the study output. In the present study, weights assigned to the experts are varied based on their work experience. Precisely, the

experts are categorized into two groups: the first group (G1) has work experience <10 years, and the second group (G2) has work experience >10 years. We consider four different scenarios, with weights of G2 ranging between 0.75 and 1.35 times of G1 with an interval of 0.20. Hence, we perform a total of four different runs under the sensitivity analysis. Table VIII provides details regarding the relative weights of the two groups of experts.

The sensitivity analysis results reveal that the elements of the cause-and-effect group remain the same in all four scenarios and are similar to the base case. Precisely, we find that LCC, FR, AWC, CH, ND, CC, and GR are cause barriers, and IDC, TP, SC, LC, NS, and SM are categorized as effect barriers in the sensitivity analysis as well as base case (Fig. 3). Moreover, the top five priority barriers ($R_i + C_j$) for all the four scenarios are "lack of GRs", SC, CC, "lack of LC," and "low FR," i.e., $GR > SC > CC > LC > FR$, which is the same as the findings of base-case (Fig. 3). These results suggest that the study is robust and free from any bias.

D. Theoretical Implications

The present study offers four key theoretical implications. First, the study provides a detailed understanding of the relationship between various barriers that affect the adoption of delivery drones in RHSC using Grey-DEMATEL. Interrelationship assessment helps highlight the logic of how the interactions result in poor adoption and the order in which these barriers should be addressed. This study is timely and necessary because of the growing human suffering and the potential of delivery drones to increase health coverage and mitigate these sufferings. While very few studies have investigated the applicability of delivery drones in RHSC, little understanding exists about the barriers that hinder their adoption in RHSC to deliver medical supplies. Therefore, the present study fills this literature gap by investigating the key barriers inhibiting the adoption of delivery drones in RHSC.

Second, the present study adds to the scarce literature on RHSC and drone delivery by incorporating barriers that were not examined in the past literature. For instance, SCs that relate to the threat to national security have not been investigated in the past. Addressing SC is necessary because terrorists and criminals may use drones to collect sensitive data, such as the location of military bases, the security level of VIPs and may even use them for explosions. Thus, by investigating this unique variable, the

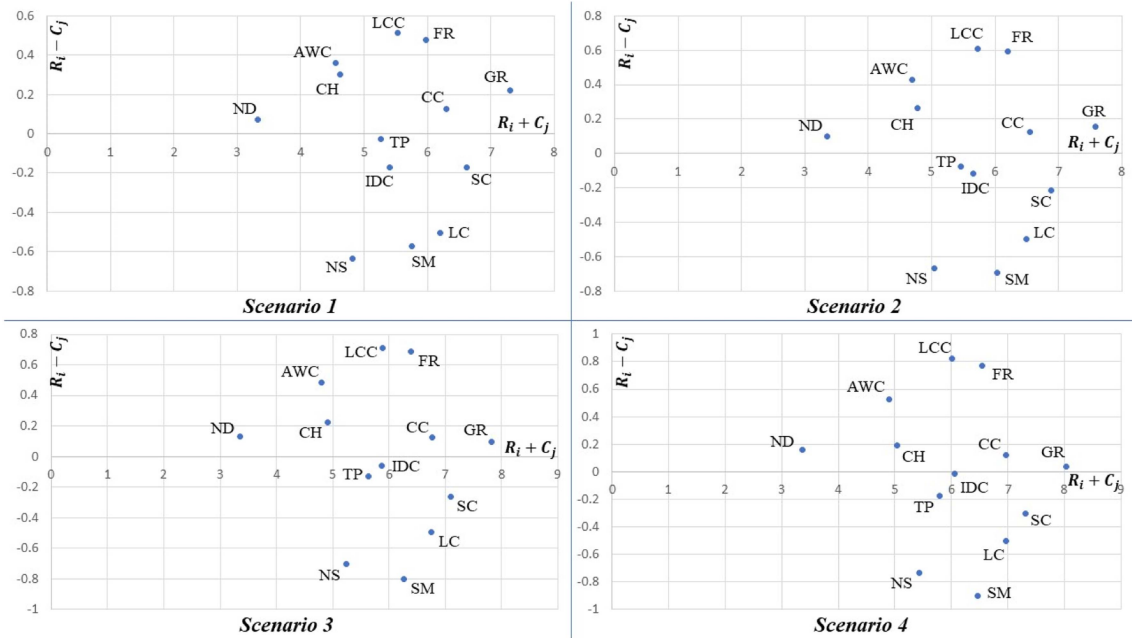


Fig. 3. Sensitivity analysis results.

study encourages future research to consider this high priority causal barrier while studying delivery drones in RHSC.

Third, the results show that majority of the barriers in the cause group are technical barriers. This finding is important because transportation scientists have majorly studied delivery drones in combination with trucks. Trucks help compensate for the technical limitations of the drones. For example, trucks could carry multiple drones and take them near the delivery point. Hence, technical barriers, such as low FR are not an issue in truck–drone delivery (used for last-mile deliveries in urban areas). However, in RHSC, drones are seen as an alternative to truck or road delivery due to poor road infrastructure and faster transportation requirements. As a key contribution, the study emphasizes upgrading delivery drone technology and incorporating technical factors while designing a drone-based RHSC network.

Finally, the study also adds to the growing body of literature on healthcare supply chain resilience. The drones may improve healthcare supply chain resilience by reducing delivery time, increasing responsiveness, and providing access to rural areas. Despite the enormous potential and ability to revolutionize the healthcare supply chains, the negative valence of delivery drones overpowers its positive valence, leading to poor adoption. Thus, the present study contributes to the literature by investigating the key barriers contributing to negative valence that hinder delivery drones' adoption in RHSC. Precisely, the present study highlights the key barriers and a systematic approach to address them, which could lead to their successful adoption in RHSC, ultimately increasing the resilience of RHSC.

E. Practical Implications

The current study's findings significantly contribute to the practice by assisting the policymakers and decision-makers

in improving health coverage and achieving SDG3. First, a two-way causal relation between “lack of GRs” and “lack of LC,” and another two-way causal relationship between “lack of GRs” and technical barriers suggest a vital role of decision-makers in policy designing and implementation. In this regard, 1) the decision-makers should provide relevant information to policymakers regarding the challenges faced in obtaining trial and testing approvals. Such information will assist policymakers in streamlining the approval process, which will promote research and development. 2) The decision-makers should inform policymakers about operational difficulties, raw material requirements, and other business challenges. These inputs will aid in developing policies that favor the availability of raw materials at a subsidized price and encourage the development of innovative business models. 3) The policymakers should be informed regarding various types of data collected by drones during operations. Such information will aid in formulating policies that protect people's privacy and the nation's security.

Second, the findings suggest that technological capabilities are a must for the smooth adoption of delivery drones in RHSC. Technical capabilities, especially long FR, higher load-carrying capacity, and obstacle avoidance, are essential from the RHSC perspective. The rural areas are thinly populated, geographically dispersed, and lack infrastructure. Therefore, delivery drones with long FRs and high load-carrying capacity are certain to be embraced by RHSC leaders. Moreover, the same fleet of technologically advanced delivery drones could assist in managing disastrous situations, such as COVID-19 [9], by flying medical essentials to the RHF as per demand to confine the pandemic growth [22]. Similarly, in the case of other disasters, such as cyclones and floods, delivery drones could substantially reduce human suffering by timely delivery of medical essentials [34].

Finally, the study underlines the importance of a skilled workforce in successfully adopting delivery drones in RHSC. Drones

rely on sophisticated technologies, such as artificial intelligence and machine learning, requiring a skilled workforce [28]. Moreover, flying drones is considered equivalent to flying an aircraft, and the crashing of drones may result in economic loss, damage to drones, injury to a person, and property damage. In this vein, the study findings suggest decision-makers to increase the availability of a skilled workforce for the sector. The decision-makers or policymakers may start with short-term training programs and work toward establishing training institutes to provide necessary skills and training to interested individuals. They should also publicize how the applicability and adoption of drones are increasing, their advantages over traditional modes of operation, and highlight the career opportunities it offers. These measures could help attract people and improve the availability of skilled labor in the sector.

VI. CONCLUSION

Governments worldwide are intensely working toward increasing health coverage and achieving SDG3. The timely delivery of medical essentials to RHF, however, remains a major challenge. In this regard, drone-based delivery is the most promising technological advancement toward this goal. Delivery drones are viewed as a truly disruptive technology in the context of RHSC due to the advantages it offers over road transport, such as low delivery time, low operational cost, and unaffected by poor road infrastructure. However, currently, the barriers seem to overpower these advantages, affecting their adoption in RHSC. To this end, we analyzed the barriers hindering the adoption of delivery drones in RHSC using the Grey-DEMATEL technique. Our results suggested that “lack of GRs” and technical barriers (“low FR,” “limited LCC,” and “risk of CC”) are the most important causal barriers that should be addressed on priority basis to accelerate the adoption of delivery drones in RHSC.

There are several interesting research questions that future work could address. First, while the present study was performed in the Indian context, a comparative cross-country analysis could shed light on different challenges and barriers at the core of the drone delivery system in other countries. Comparing developing and developed countries is essential as developed countries are early adopters, while developing countries follow them. Therefore, such research could help transfer best practices and barrier mitigation strategies from developed to developing countries. Second, the current study is limited to the use of delivery drones in RHSC. Small drones, on the other hand, may have applications inside hospitals. These applications could include spraying and sanitation, delivering medicines to patients, transporting equipment from one floor to another, and temperature monitoring [9]. Future work could assess the challenges to drone technology adoption for these purposes using the framework suggested in the present study. Such research will help expand the use of drones in the healthcare sector and improve the resilience of healthcare supply chains. Third, while the truck–drone combination for e-commerce or retail delivery has gained considerable attention from transportation scientists, the studies on delivery drones in RHSC are still scarce. Therefore, future research should explore network design problems using delivery

drones in the RHSC context. Moreover, the key technical barriers should be incorporated as constraints in these network design problems to make the model more practically relevant. Fourth, the study paves way for empirical assessment in this domain. Diverse sample sets originating from contextually different RHSCs could be examined to understand, evaluate, and quantify the impact of barriers highlighted in the present study. The results of such studies should be compared with findings of the current study. In addition, empirical examinations assessing the role of delivery drones in enhancing the supply chain responsiveness could be studied.

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